Dibendu Singh

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Professional Summary

Ph.D. candidate in Chemistry specializing in structural biology and molecular motor proteins, with a focus on DNA translocase SpoIIIE. Experienced in cryo-EM data processing (CryoSPARC), protein purification, and enzymatic kinetics (ATPase, EMSA). Investigating the mechanistic relationship between oligomerization, structure, and ATPase activity in DNA translocation systems.

Education

PhD in Chemistry — *July 2024 – June 2029 (expected)*
Indiana University Bloomington

MS in Chemistry — *July 2022 – June 2024*
Indian Institute of Technology Bombay

BS in Chemistry — *July 2019 – June 2022*
University of Calcutta

Technical Skills

Structural Biology: Cryo-EM data processing and single-particle analysis (CryoSPARC), model building (ChimeraX, PyMOL)
Biochemistry: Recombinant DNA Technology, Gene Cloning, Site-Directed Mutagenesis, Protein Expression (E. coli, BL21 Star).
Purification: FPLC (AKTA), Affinity Chromatography (Ni-NTA), Size Exclusion (SEC), Ion Exchange.
Software/Tools: CryoSPARC, ChimeraX, PyMOL, Python, Shell Scripting (Linux/macOS), LaTeX.

Research Projects

Graduate Student — *Indiana, IN*
Marc C. Morais Lab, Indiana University Bloomington — *August 2024 – present*

- Investigating the molecular mechanism of DNA translocation in SpoIIIE using integrated structural and biochemical approaches
- Purified recombinant SpoIIIE and identified higher-order oligomer (650 kDa) using SEC, indicating stable multimer formation
- Measured ATPase activity across varying protein concentrations and observed loss of cooperative behavior at lower concentration, suggesting oligomerization-dependent catalysis
- Processed cryo-EM datasets in CryoSPARC (motion correction, 2D classification, 3D reconstruction), identifying filamentous assemblies distinct from expected ring structures
- Optimized purification and assay conditions to improve protein stability and reproducibility of higher-order assemblies

MS Project — *Mumbai, India*
Prof. Ishita Sengupta, IIT Bombay — *July 2023 – June 2024*

- Developed NanoLuc-based luminescent biosensors and quantified detection sensitivity for OPS using luminescence assays
- Performed molecular cloning (homologous recombination), DNA prep/analysis, and protein purification
- Characterized protein folding (CD) and evaluated biosensor performance using luminescence assays

Summer Internship — *Pune, India*
Prof. Alok Das, IISER Pune — *May 2023 – July 2023*

- Studied intramolecular interactions (N–H$\cdots$N(C5), N–H$\cdots$S(C6)) using gas-phase laser spectroscopy and solution-phase techniques
- Performed quantum mechanical calculations and geometry optimizations of a tri-peptide molecule using GAUSSIAN